1- CLASS TASKS SOLUTIONS

--- TASK 1 SOLUTION KEY ---

1. Metrics required: Total team payroll expenditure ($) and average base salary ($).
2. Total payroll alone is insufficient because it does not reveal whether individual salaries align with company compensation bands or if a single high salary distorts the overall department budget.

--- TASK 2 & 3 SOLUTION KEY (PYTHON CODE) ---

```python
# Task 2 Setup

company_benchmark = 75000.0
proposed_salaries = [72000.0, 81000.0, 75000.0, 69000.0, 88000.0]
employee_names = ["Maya", "Liam", "Sophia", "Noah", "Ethan"]

print("Benchmark:", company_benchmark) print("Salaries:",
proposed_salaries) print("Employees:", employee_names)

# Task 3 Calculations

total_payroll = sum(proposed_salaries) team_size = len(proposed_salaries)
average_salary = total_payroll / team_size

salary_variances = [
proposed_salaries[0] - company_benchmark, proposed_salaries[1] -
company_benchmark, proposed_salaries[2] - company_benchmark,
proposed_salaries[3] - company_benchmark, proposed_salaries[4] -
company_benchmark

]

print("Total Payroll:", total_payroll) print("Team Size:", team_size)
print("Average Salary:", average_salary) print("Salary Variances:",
salary_variances) 0.0, -6000.0, 13000.0]

# Maya: -3000.0 # Liam: +6000.0 # Sophia: 0.0
# Noah: -6000.0 # Ethan: +13000.0

# Output: 385000.0
# Output: 5
# Output: 77000.0
# Output: [-3000.0, 6000.0, 0.0, -6000.0, 13000.0]
```

--- TASK 4 SOLUTION KEY (BUSINESS MEMO) ---

1. The average salary is $77,000.00, which sits $2,000.00 (+2.67%) above the $75,000.00 standard benchmark.
2. Ethan creates the largest positive variance (+$13,000.00 above benchmark).
3. Sample Recommendation Memo:

"**To**: Sarah Jenkins, HR Operations Manager **Re**: Analytics Team Salary Review

The proposed salary package for the 5-member analytics team totals $385,000, resulting in an average base salary of $77,000. While this is $2,000 per employee

above our $75,000 baseline, the variance is almost entirely driven by Ethan's senior compensation rate ($88,000). Because entry-level rates for Noah ($69,000) and Maya ($72,000) absorb part of this premium, overall departmental variance remains within a manageable threshold. I recommend approving the package."

2- HOMEWORK SOLUTION

```python

```
Homework Problem 1 & 2 Python Solution

```
target_benchmark = 20.0
overtime_hours = [14.5, 22.0, 18.5, 31.0, 12.0, 25.5, 19.0, 28.0, 15.0,
21.0]
worker_names = ["Worker_A", "Worker_B", "Worker_C",

"Worker_F", "Worker_G", "Worker_H",

total_overtime = sum(overtime_hours)
avg_overtime = total_overtime / len(overtime_hours)

var_worker_a = overtime_hours[0] - target_benchmark var_worker_b =
overtime_hours[1] - target_benchmark var_worker_c = overtime_hours[2] -
target_benchmark

"Worker_D", "Worker_E", "Worker_I", "Worker_J"]

print("Total Overtime:", total_overtime) # Output: 206.5 print("Average
Overtime:", avg_overtime) # Output: 20.65 print("Worker A Variance:",
var_worker_a)
print("Worker B Variance:", var_worker_b)

print("Worker C Variance:", var_worker_c)
```

- **Homework Business Memo Grading Criteria:**
  - Correctly identifies average overtime as 20.65 hours (0.65 hours above target).
  - Notes that overtime is unevenly distributed (e.g., Worker_D logged 31.0 hours; Worker_H logged 28.0 hours).
  - Recommends investigating high-hour workers for burnout risks and redistribution of shift tasks.